

18_Cramer's Rule

What is Cramer's Rule?

Cramer's Rule is a method in **linear algebra** used to solve a system of **linear equations** when:

- The number of equations = number of unknowns
- The determinant of the coefficient matrix is **non-zero**

General Form (2 variables)

Given a system:

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$$

Step 1: Determinant of coefficients

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = a_1b_2 - a_2b_1$$

If $D \neq 0$, a unique solution exists.

Step 2: Determinants for x and y

$$D_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix} = c_1 b_2 - c_2 b_1$$
$$D_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix} = a_1 c_2 - a_2 c_1$$

Step 3: Solution

$$x = \frac{D_x}{D}, y = \frac{D_y}{D}$$

Example Problem

Solve:

$$\begin{cases} 2x + 3y = 11 \\ 4x - y = 5 \end{cases}$$

Step 1: Coefficient determinant

$$D = \begin{vmatrix} 2 & 3 \\ 4 & -1 \end{vmatrix} = (2)(-1) - (4)(3) = -2 - 12 = -14$$

Step 2: D_x

$$D_x = \begin{vmatrix} 11 & 3 \\ 5 & -1 \end{vmatrix} = (11)(-1) - (5)(3) = -11 - 15 = -26$$

Step 3: D_y

$$D_y = \begin{vmatrix} 2 & 11 \\ 4 & 5 \end{vmatrix} = (2)(5) - (4)(11) = 10 - 44 = -34$$

✓ Final Solution

$$x = \frac{-26}{-14} = \frac{13}{7}, y = \frac{-34}{-14} = \frac{17}{7}$$

Consider a system of three linear equations:

$$\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases}$$

Step 1: Form the coefficient matrix

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

Compute its determinant:

$$\Delta = \det(A)$$

⚠ Cramer's Rule works **only if** $\Delta \neq 0$

Step 2: Determinants for each variable

Replace one column at a time with the constants d_1, d_2, d_3 :

$$\Delta_x = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$$

$$\Delta_y = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$$

$$\Delta_z = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

Step 3: Solve

$$x = \frac{\Delta_x}{\Delta}, y = \frac{\Delta_y}{\Delta}, z = \frac{\Delta_z}{\Delta}$$

Worked Example (3 Variables)

Solve:

$$\begin{cases} 2x + y - z = 8 \\ -3x - y + 2z = -11 \\ -2x + y + 2z = -3 \end{cases}$$

Step 1: Determinant of coefficient matrix

$$\Delta = \begin{vmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{vmatrix}$$

$$\Delta = 2((-1)(2) - (2)(1)) - 1((-3)(2) - (2)(-2)) + (-1)((-3)(1) - (-1)(-2))$$

$$\Delta = 2(-4) - 1(-2) + (-1)(-5)$$

$$\Delta = -8 + 2 + 5 = -1$$

Step 2: Determinants for x, y, z

Calculating Δ_x

$$\Delta_x = \begin{vmatrix} 8 & 1 & -1 \\ -11 & -1 & 2 \\ -3 & 1 & 2 \end{vmatrix}$$

Expand along the first row:

$$\Delta_x = 8 \begin{vmatrix} -1 & 2 \\ 1 & 2 \end{vmatrix} - 1 \begin{vmatrix} -11 & 2 \\ -3 & 2 \end{vmatrix} + (-1) \begin{vmatrix} -11 & -1 \\ -3 & 1 \end{vmatrix}$$

Compute each 2×2 determinant:

$$\begin{vmatrix} -1 & 2 \\ 1 & 2 \end{vmatrix} = (-1)(2) - (2)(1) = -4$$

$$\begin{vmatrix} -11 & 2 \\ -3 & 2 \end{vmatrix} = (-11)(2) - (2)(-3) = -16$$

$$\begin{vmatrix} -11 & -1 \\ -3 & 1 \end{vmatrix} = (-11)(1) - (-1)(-3) = -14$$

Substitute back:

$$\Delta_x = 8(-4) - 1(-16) + (-1)(-14)$$

$$\Delta_x = -32 + 16 + 14 = \boxed{-2}$$

Calculating Δ_y

$$\Delta_y = \begin{vmatrix} 2 & 8 & -1 \\ -3 & -11 & 2 \\ -2 & -3 & 2 \end{vmatrix}$$

Expand along the first row:

$$\Delta_y = 2 \begin{vmatrix} -11 & 2 \\ -3 & 2 \end{vmatrix} - 8 \begin{vmatrix} -3 & 2 \\ -2 & 2 \end{vmatrix} + (-1) \begin{vmatrix} -3 & -11 \\ -2 & -3 \end{vmatrix}$$

Compute determinants:

$$\begin{vmatrix} -11 & 2 \\ -3 & 2 \end{vmatrix} = -22 + 6 = -16$$

$$\begin{vmatrix} -11 & 2 \\ -3 & 2 \end{vmatrix} = (-3)(2) - (2)(-2) = -2$$

$$\begin{vmatrix} -11 & 2 \\ -3 & 2 \end{vmatrix} = 9 - 22 = -13$$

Substitute:

$$\Delta_y = 2(-16) - 8(-2) + (-1)(-13)$$

$$\Delta_y = -32 + 16 + 13 = \boxed{-3}$$

Calculating Δ_z

$$\Delta_z = \begin{vmatrix} 2 & 1 & 8 \\ -3 & -1 & -11 \\ -2 & 1 & -3 \end{vmatrix}$$

Expand along the first row:

$$\Delta_z = 2 \begin{vmatrix} -1 & -11 \\ 1 & -3 \end{vmatrix} - 1 \begin{vmatrix} -3 & -11 \\ -2 & -3 \end{vmatrix} + 8 \begin{vmatrix} -3 & -1 \\ -2 & 1 \end{vmatrix}$$

Compute determinants:

$$\begin{vmatrix} -1 & -11 \\ 1 & -3 \end{vmatrix} = 3 + 11 = 14$$

$$\begin{vmatrix} -3 & -11 \\ -2 & -3 \end{vmatrix} = 9 - 22 = -13$$

$$\begin{vmatrix} -3 & -1 \\ -2 & 1 \end{vmatrix} = -3 - 2 = -5$$

Substitute:

$$\Delta_z = 2(14) - 1(-13) + 8(-5)$$

$$\Delta_z = 28 + 13 - 40 = \boxed{1}$$

Step 3: Final Answer

$$x = \frac{-2}{-1} = 2$$

$$y = \frac{-3}{-1} = 3$$

$$z = \frac{1}{-1} = -1$$

 **Solution:**

$$\boxed{x = 2, y = 3, z = -1}$$